

AST1410 MESA Practical 2

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For consistency (and because my file organization was very bad for the first practical), the original simulation was rerun up to the post MS turnoff criterion of minimum core H1 fraction 10^{-3} alongside the two new stars with `mixing_length_alpha`, which we call α , set to 1 and 4 instead of the default 2. The evolutionary trajectories of all three stars are shown in the same plots below. The main modifications made to the files from last time were: the removal of `inlist_load`; the reinstatement of the stopping condition mentioned above; the specification of the `mixing_length_alpha` parameter; additions to `plot.py`, which is used to generate all the plots. All files are uploaded to the main branch on https://github.com/Gursach44/MESA_practicals/tree/main/MESA_practical_2.

Overall, increasing α appears to have several effects. Figure 1 demonstrates that larger values of α result in higher effective temperatures for the star at ZAMS. This indicates energy is more easily transported to the surface, which is consistent with a higher convective efficiency. Figures 2 and 3 exhibit a higher initial central density for stars with higher α as well; this roughly implies the star is more compact. This makes sense since improved energy transport to the surface would result in the star being less ‘puffed up’ for a given energy generation rate internally. Notably, once on the main sequence, all stars appear to converge in their evolutionary trajectories. This also aligns with expectation; since we are modelling high mass ($15M_{\odot}$) stars, we expect them to have deep radiative envelopes, meaning the dependence of radiation transport on convective parameters weakens as the star ages.

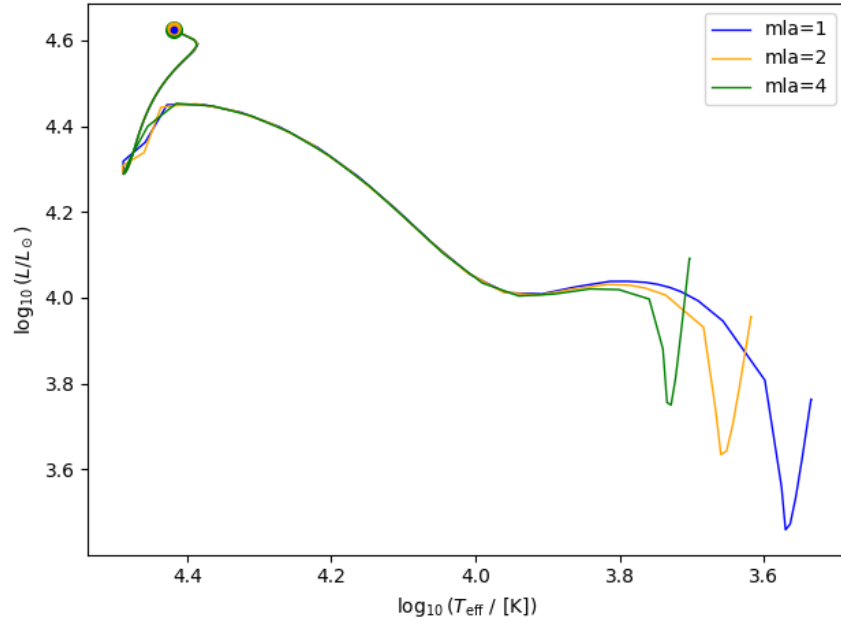


Figure 1: HR diagram showing evolution of 15 solar mass stars with varying `mixing_length_alpha` parameter values in MESA from slightly pre-MS to slightly post-MS. Endpoints of each evolutionary track are marked with dots of the same colours as the lines.

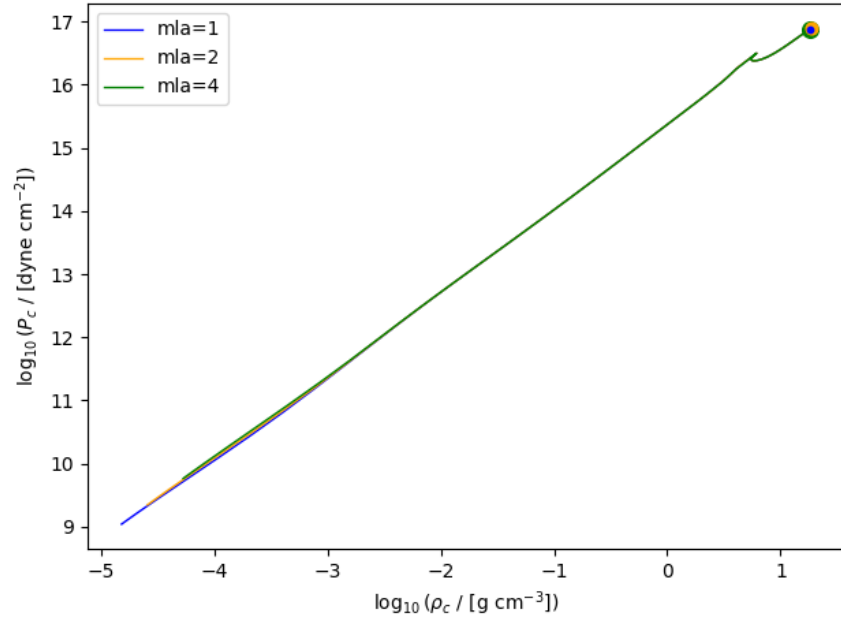


Figure 2: $\log P_c$ vs $\log \rho_c$ plot showing evolution of 15 solar mass stars with varying `mixing_length_alpha` parameter values in MESA from slightly pre-MS to slightly post-MS. Endpoints of each evolutionary track are marked with dots of the same colours as the lines.

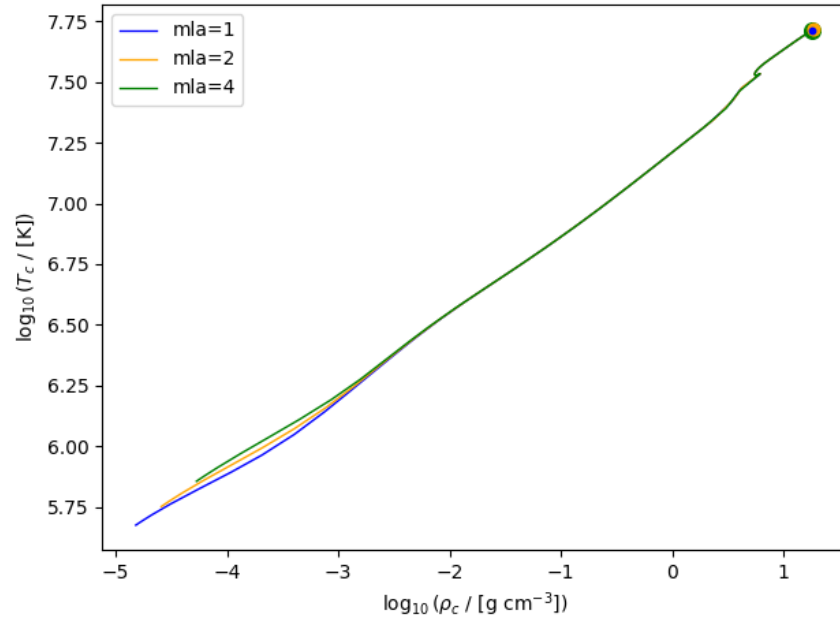


Figure 3: $\log T_c$ vs $\log \rho_c$ plot showing evolution of 15 solar mass stars with varying `mixing_length_alpha` parameter values in MESA from slightly pre-MS to slightly post-MS. Endpoints of each evolutionary track are marked with dots of the same colours as the lines.